Stanford Anatomic Pathology
Cytopathology Service
Section Director: [redacted]
Director of Anatomic Pathology: [redacted]

Addended Report: AP FNA

Bindal, Yashdeep

[redacted]

PATIENT NAME	DOB	MRN	SEX
Bindal, Yashdeep	[redacted]	[redacted]	Male

FNA (Final result) [redacted]

Authorizing Provider:	[redacted]	Ordering Provider:	[redacted]
Ordering Location:	CATH INTRA OP	Collected:	01/15/2025 1039
Pathologist:	[redacted] PhD	Received:	01/15/2025 1134

Specimens

A Liver, Liver Cores

Addendum

This addendum is to report the Claudin-18 IHC result as below. Targeted NGS testing also has been initiated and the results will be reported in Epic.

Addendum Diagnosis:

A. Liver, core biopsy

- **Limited Claudin-18.2 expression by immunohistochemistry (2-3+, 50%; see comment)**

[redacted] **PhD**

Comment:

The FDA-approved Claudin-18 assay uses antibody clone 43-14A on the Ventana Ultra platform with OptiView detection. The accepted scoring scheme when evaluating Claudin-18 includes both intensity of membranous staining and percentage of tumor cells stained. Tissue slides with tumor present can demonstrate varying levels of Claudin-18 membranous staining intensity ranging from no staining to strong staining (0 to 3+). The reported percentage of positive tumor cells only includes cells demonstrating membranous staining. Claudin-18 positivity has been defined as 75% or more of tumor cells showing moderate to strong (2-3+) membranous staining (Shitara et al., Lancet, 2023). A minimum of 50 viable tumor cells are required for evaluation.

Addendum electronically signed by [redacted] PhD on 1/21/2025 at 2104

Addendum

This case was also reviewed by Dr. [redacted] (GI Pathology), who concurs.

Claudin 18 and STAMP are pending and the findings will be reported in addenda or separately.

[redacted]

Addendum electronically signed by [redacted] on 1/17/2025 at 1805

Diagnosis

A. Liver, CT-Guided Fine Needle Aspiration and Core Biopsy

- **Involved by adenocarcinoma (see comment)**
- **Intact expression of mismatch repair (MMR) proteins by immunohistochemistry**
- **Negative for HER2 (0) by immunohistochemistry**

[redacted]

Electronically signed by [redacted] on 1/17/2025 at 1751

Diagnosis Comments

The morphologic features are diagnostic of adenocarcinoma. While the immunophenotype is non-specific, it is compatible with a pancreatobiliary or upper GI primary. Correlation with clinical and imaging findings is recommended. Preliminary findings discussed with Dr. [redacted] via telephone by Dr. [redacted] on 01/16/25 at 5:45 PM. Final results communicated with Dr. [redacted] by Dr. [redacted] on 01/17/25 at 12:34 PM via secure message.

This case was also reviewed by Dr. [redacted] (Cytopathology), who concurs.

Microscopic Description

Cytologic preparations demonstrate hypercellular smears consisting of crowded cohesive sheets of malignant epithelial cells displaying marked anisonucleosis (>4:1), ovoid/irregular to columnar nuclei, prominent nucleoli, and moderate to abundant amounts of delicate cytoplasm. The core biopsies show infiltrative malignant glands involving fibrous tissue.

Ancillary Studies

Ancillary studies have been performed on block A2, which contains approximately 60-70% tumor with the following immunophenotype:

Task ID	Task Name	Result
A2-2	KERATIN7 (OV-TL 12/30) IHC	Positive
A2-3	KERATIN20 (Ks20.8) IHC	Positive, focal
A2-4	MLH1 (G168-728) IHC	Intact
A2-5	MSH2 (FE11) IHC	Intact
A2-6	MSH6 (44) IHC	Intact
A2-7	PMS2 (MRQ-28) IHC	Intact
A2-8	HER2 (4B5) IHC	Negative (0)
A2-9	SATB2 (EP281) IHC	Positive, subset, weak
A2-10	BETACATENIN (14) IHC	Negative (membranous staining pattern)

Clinical History

Pancreatic tail mass, numerous liver lesions, and FDG avid intrathoracic and intraabdominal lymph nodes noted on imaging.

Procedure Note/Gross Description

A. Liver.

Cytology Materials Collected:

3Air-dried slide(s) and 1Alcohol-fixed slide(s)
Multiple visibly adequate red/tan tissue cores (1.0 x 0.1 x 0.1cm) for cell block

Immediate Evaluation

A. Liver.

Representative Diff-Quik stained aspirate smears were reviewed for immediate evaluation by [redacted] and Dr.

[redacted] via Spot webcam:
Evaluation Episode #1: Adequate

Disclaimer

The microscopic interpretation(s) on this report have been rendered through the microscopic review of glass slides or use of whole slide imaging (digital pathology). Any immunologic tests performed on this case were developed and its performance characteristics determined by Stanford Health Care Immunoperoxidase Laboratory. Unless indicated otherwise, the assay was performed on formalin-fixed, paraffin-embedded tissue with polymer detection system. It has not been cleared or approved by the USFDA, although such approval is not required for analyte-specific reagents of this type. This assay has not been validated on decalcified tissues. Results should be interpreted with caution given the possibility of false negativity on decalcified specimens. All controls show appropriate reactivity.

Grossing Location

Stanford Anatomic Pathology and Clinical Laboratories

CC List

[redacted]

Resulting Lab(s)

Laboratory Name

Stanford Anatomic Pathology and Clinical Laboratories

Stanford Clinical Laboratory at Hillview

Stanford Health Care Emeryville Clinical Laboratory

Stanford Health Care Laboratory South Bay

Stanford Health Care Tri-Valley

[redacted]